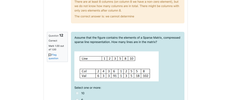

Data Structures and Algorithms - 2025 - 2026

Quiz navigation

1 ✓ 2 ✓ 3 ✓ 4 ✓ 5 ⊗
6 ✓ 7 ✓ 8 ◐ 9 ✓ 10 ⊗
11 ⊗ 12 ✓ 13 ✓ 14 ⊗ 15 ✓
16 ⊗

Show one page at a time
Finish review

Status	Finished
Started	Wednesday, 25 March 2026, 11:16 AM
Completed	Wednesday, 25 March 2026, 11:43 AM
Duration	27 mins 39 secs
Marks	10.50/16.00
Grade	**6.56** out of 10.00 (**65.63**%)

Question **1**
Correct
Mark 1.00 out of 1.00
Flag question

Choose the statements that are true for ADT Bag.

Select one or more:

- ☐ It contains key-value pairs
- ☑ It does not have positions ✓
- ☐ It has unique elements
- ☐ Elements have positions
- ☐ It does not have an iterator
- ☑ The same element can occur multiple times ✓
- ☐ It cannot be implemented on a Dynamic Array

Your answer is correct.

In a Bag we can have the same element multiple times and the order of the elements is not important.

The correct answers are: It does not have positions, The same element can occur multiple times

Question **2**
Correct
Mark 1.00 out of 1.00
Flag question

For representation R3 for the Bag (discussed at Lecture 3) we talked about how remove is a more complex operation. In order to make it easier, assume that we do the following modification:

to every element in the elems array (the one holding the unique elements) we add a boolean flag, called "unique" which is true if the element appears only once in the Bag and False otherwise. Positions array is defined just like in case of R3.

For example a Bag with elements 4, 1, 6, 4, 7, 2, 1, 1, 1, 9 would have the following representation (first array is elems with the boolean flag, second array is positions):

1	2	3	4	5	6	7	8	9
4, F	1, F	6, T	7, T	2, T	9, T			

1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	2	3	1	4	5	2	2	2	6				

Which statements are true about the operations for this new representation?

- ☐ Adding an element will have a higher total time complexity.
- ☐ Removing an element will have a lower total time complexity, since we do not need to check if this was the last occurrence of the element.
- ☐ Removing an element will have a higher total time complexity, since we need to update the boolean flags.
- ☐ Returning the number of occurrences of a given element will have a lower total time complexity.
- ☑ Nothing changes in the time complexity of operations. ✓

Your answer is correct.

Adding the boolean flags will not change the complexity of the operations. Let's see what extra work we have due to the flags:

- add - if we add an element for the first time, we need to set its flag to T. If we add an element for the second time, we need to change its flag from T to F, and if an element that we add already has the flag set to F we do not change it. But add starts anyway with a traversal of the elems array, to see if the element is there and on what position (remember, its position is what we put in the positions array). Changing the flag is just one extra operation, which does not change the complexity.

- remove - When we remove an element we first go through the elems array to find its position. Then, we go through the positions array to find the element and remove it. Then we need to continue searching the positions array to see if the element is there multiple times (and if it does not appear anymore, we have a set of other things to do). This is where the boolean flag helps, if the element that we wanted to remove had a flag T (so it was unique) we know that it will not be present in the positions array any more time. However, assume that an element appears twice in the bag, and you remove one occurrence. Now its flag should be set from F to T (element became unique). So, whenever you remove an element with a F flag, you need to go through the positions array to check if the element appears any more time, otherwise its flag needs to be set to T.

- remove - The same as above, the complexity will not increase, just the worst case situation is only when the flag is False, not every time when an element is found.

- nothing changes for number of occurrences, elements with flag T have an occurrence of 1, but for elements with flag F we still need to go through the positions array to see how many times they occur.

The correct answer is:
Nothing changes in the time complexity of operations.

Question **3**
Correct
Mark 1.00 out of 1.00
Flag question

Consider a Bag in representation R3 from Lecture 2 (see figure below for example). Assume that you want to implement an iterator for it. What fields / attributes / variables (pick the term which you understand better) should be part of the representation of the iterator? Assume that everything you choose will be part of the iterator together.

	1	2	3	4	5	6	7	8	9
elems	4	1	6	7	2	9			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
positions	1	2	3	1	4	5	2	2	2	6				

- ☑ the bag ✓ Every iterator needs to contain a reference to the container it iterates over. An iterator for a bag, needs to contain the bag.
- ☐ an integer value, denoting the current position in the elems array
- ☑ an integer value, denoting the current position in the positions array ✓ We need this, this is how you know your current element. The only difference compared to a regular dynamic array iterator, is that in this case the element from this current position is not directly your element from the bag, only its position from the elems array.
- ☐ a boolean flag to show when the iterator is invalid
- ☐ an integer value denoting the frequency of the current element

Your answer is correct.

Every iterator needs to contain a reference of the container it iterates over, so we need the bag.

Iteration will happen through the positions array, so we need a current position there. We do not need a current position for the elems array, basically, what we have in the positions array needs to be interpreted as position in that array.

Since we have the length of the positions array, we do not need a boolean flag to know when the iterator is invalid.

Since this is not a frequency based representation, we do not need current frequency.

The correct answers are: the bag,
an integer value, denoting the current position in the positions array

Question **4**
Correct
Mark 1.00 out of 1.00
Flag question

From the previous question we know that an Iterator for a Bag in representation R3 should have the following representation:

BagIterator:

b: Bag //contains 2 dynamic arrays, elems and positions

currentPosition: Integer // current position in the positions dynamic array

Consider the following alternative representation for the iterator:

BagIterator2:

b: Bag // contains 2 dynamic arrays, elems and positions

currentPosition: Integer // current position in the elems dynamic array

currentFrequency: Integer // the current frequency of the element from the currentPosition

Assume that you intend to implement the BagIterator2 in the following way: instead of iterating through the positions dynamic array, you will iterate over the elems array, this is why you have the currentPosition in that array. But to account for frequencies (since elems contains unique elements) each time the currentPosition changes, we go through the positions dynamic array and count how many occurrences that element actually has in the Bag. We put this value in currentFrequency.

Then the iterator operations would be implemented in the following way:

Next: if currentFrequency is zero, incremenet currentPosition and reset the value of currentFrequency to be the number of occurrences of the new position in the positions array. If currentFrequency is not zero, decrement currentFrequency and do not change anything else.

getCurrent: simply return the element from the elems array from position currentPosition

valid: check if currentPosition is less than or equal to the size of the elems dynamic array and that currentFrequency is greater than 0.

Which statement is true about BagIterator and BagIterator2?

- ☐ Both of them are correct (they return all the elements of the Bag), and they are equally efficient.
- ☑ Both of them are correct (they return all the elements of the Bag), but BagIterator2 is less efficient. ✓
- ☐ BagIterator2 is not correct, it would not return all the elements from the Bag
- ☐ BagIterator2 is not correct, it returns all elements, but in the wrong order

Your answer is correct.

The above presented next operation from BagIterator2 will take time Theta(positions.size) to count the frequencies of an element, not Theta(1) as we would like from an iterator (and which is possible with BagIterator).

The correct answer is:
Both of them are correct (they return all the elements of the Bag), but BagIterator2 is less efficient.

Question **5**
Incorrect
Mark 0.00 out of 1.00
Flag question

You can see below the content of a Bag where R4 was used as representation (positions represent the element and values from positions are the frequencies). Which of the following statements is true about this Bag?

4	0	1	0	0	0	2	0	6	0	0	0	1

Select one or more:

- ☐ The representation is not complete
- ☐ There are 14 elements in the Bag.
- ☑ Number 6 is in the bag. ⊗ False. We have a 6 in the array, but that is a frequency, so there is an element which appears 6 times. But we do not know which, because we do not know the minimum element.
- ☐ There are 13 different/unique elements in the Bag.

Your answer is incorrect.

In order to interpret/understand this representation, we need to know the minimum element, this is going to tell us, which element's frquency can be found on the first position -> in the current form the representation is not complete.

However, while not knowing the exact elements, we know that we have 5 unique elements and 14 elements in total. (and that the difference between the minimum and maximum element is 12).

The correct answers are:
The representation is not complete,

There are 14 elements in the Bag.

Question **6**
Correct
Mark 1.00 out of 1.00
Flag question

Choose the statement(s) that are true for ADT Set.

Select one or more:

- ☑ It can have an iterator ✓
- ☐ It contains key-value pairs
- ☐ The same element can appear multiple times
- ☑ Elements do not have positions ✓
- ☑ Elements have to be unique ✓
- ☐ It cannot be implemented on a dynamic array

Your answer is correct.

Set is a container in which the elements are unique and they have no position

The correct answers are: It can have an iterator, Elements do not have positions, Elements have to be unique

Question **7**
Correct
Mark 1.00 out of 1.00
Flag question

You can see below the content of a Set of integer numbers, represented on a dynamic array of boolean values. If we implement an iterator over this Set, which operation will not have a complexity of $\Theta(1)$?

T	F	F	F	T	F	F	F	T	F	F	F	F	F	T

Minimum element: -5

Select one or more:

- ☐ init
- ☐ getCurrent
- ☑ next ✓
- ☐ valid
- ☐ none of them

Your answer is correct.

Current element of the iterator will be a position in the array. However, the iterator goes through the existing elements only, so it has to skip the False values. This is why the next operation is not just currentElement++, we have to keep on going until we find a True or get to the end of the array.

The correct answer is: next

Question **8**
Partially correct
Mark 0.50 out of 1.00
Flag question

Assume this is the content of a Set of integer numbers, where a boolean array is used for representation. Which of the following elements are in the Set?

T	F	F	F	T	F	F	F	T	F	F	F	F	F	T

Minimum element: -5

Select one or more:

- ☐ 0
- ☑ 3 ✓
- ☐ -5
- ☐ 10
- ☐ 1

Your answer is partially correct.

You have correctly selected 1.
The first element is the minimum (-5). The existing elements are the ones where we have a True.

The correct answers are: 3, -5

Question **9**
Correct
Mark 1.00 out of 1.00
Flag question

The interface of ADT Bag and ADT Set is pretty similar. What operation(s) from the interface of the Bag do(es) not exist for a Set?

Select one or more:

- ☐ add
- ☐ remove
- ☐ size
- ☑ nrOccurrences ✓
- ☐ iterator
- ☐ search

Your answer is correct.

In a Set we have unique elements, so having an operation nrOccurrences makes no sense (if an element is in the set its occurrence is 1).

The correct answer is: nrOccurrences

Question **10**
Incorrect
Mark 0.00 out of 1.00
Flag question

Assume that the following figure contains the content of a Sparse Matrix, representation where <line, column, value> triples are kept. How many columns does the Matrix have?

Line	1	1	3	3	4	4
Col	1	7	5	9	4	5
Val	11	9	30	7	20	39

Select one or more:

- ☐ 6
- ☑ 9 ⊗
- ☐ 5
- ☐ 10
- ☐ we cannot determine

Your answer is incorrect.

We know that there are at least 9 columns, since on column 9 we have a non-zero element. But if there are columns at the end of the matrix with only zero elements, they are not visible on this representation.

The correct answer is: we cannot determine

Question **11**
Incorrect
Mark 0.00 out of 1.00
Flag question

Assume that the figure below contains a Sparse Matrix in compressed sparse line representation. How many columns are in the matrix?

Line	1	2	3	5	8	10

Col	2	4	3	6	1	2	5	5	8
Val	6	3	3	91	1	3	5	18	102

Select one or more:

- ☐ 9
- ☑ 8 ⊗
- ☐ 10
- ☐ we cannot determine

Your answer is incorrect.

There are at least 8 columns (on column 8 we have a non-zero element), but we do not know how many columns are in total. There might be columns with only zero elements after column 8.

The correct answer is: we cannot determine

Question **12**
Correct
Mark 1.00 out of 1.00
Flag question

Assume that the figure contains the elements of a Sparse Matrix, compressed sparse line representation. How many lines are in the matrix?

Line	1	2	3	5	8	10

Col	2	4	3	6	1	2	5	5	8
Val	6	3	3	91	1	3	5	18	102

Select one or more:

- ☐ 10
- ☐ 6
- ☑ 5 ✓
- ☐ we cannot determine

Your answer is correct.

In compressed sparse line representation the line array has always number of lines + 1 elements. Since in the example it contains 6 elements, there are 5 lines in the matrix.

The correct answer is: 5

Question **13**
Correct
Mark 1.00 out of 1.00
Flag question

For ADT Map, what is the parameter for operation search (besides the map), and what does the operation return?

Select one or more:

- ☐ The parameter is a key and a value and the operations returns true or false depending on whether the pair is in the map or not.
- ☐ The parameter is a key and the operation returns true or false depending on whether the key is in the map or not.
- ☑ The parameter is a key and the operation returns the value associated to this key or null_tvalue if the key is not in the map. ✓
- ☐ The parameter is a value and the operation returns true or false depending on whether this value is in the map or not.
- ☐ The parameter is a value and the operation returns the key associated to this value or null_tkey if the value is not in the map.

Your answer is correct.

While for most containers search is a boolean operation, here it is different. Operations in a Map happen based on a key. So search receives as parameter the key and returns the associated value.

The correct answer is: The parameter is a key and the operation returns the value associated to this key or null_tvalue if the key is not in the map.

Question **14**
Incorrect
Mark 0.00 out of 1.00
Flag question

What is the main difference between ADT Set and ADT Map?

Select one or more:

- ☑ In a Set elements are unique, but in a Map we can have the same element multiple times. ⊗
- ☐ In a Map elements are unique, but in a Set we can have the same element multiple times.
- ☐ A Map can be implemented on a dynamic array, but a Set cannot.
- ☐ A Set can have an iterator, but a Map cannot.
- ☑ A Map contains key-value pairs while a Set contains simple elements. ✓
- ☐ A Set does not have positions, but a Map does.
- ☑ A Map does not have positions, but a Set has. ⊗

Your answer is incorrect.

ADT Set and ADT Map are pretty similar, none of them have positions, both can have iterators, and both contain unique elements. The main difference is that in a Set we have simple elements, while in a Map we have key-value pairs (and actually the keys are unique).

The correct answer is: A Map contains key-value pairs while a Set contains simple elements.

Question **15**
Correct
Mark 1.00 out of 1.00
Flag question

ADT Map and ADT MultiMap have a pretty similar interface. Which of the following operations do not have the same number of input parameters for the two?

Select one or more:

- ☐ add
- ☑ remove ✓
- ☐ size
- ☐ iterator
- ☐ isEmpty

Your answer is correct.

For add, they both receive a key and a value.

For search, they both receive a key and return a value/list of values. (Obs. in the interface of the MultiMap this list of values appears as an output parameter, but we are not counting it, since the questions asked for input parameters).

Size, iterator and isEmpty take no parameter, besides the container.

In case of a Map, since keys are unique, when we remove an element, it is enough to specify its key. But in a Multimap a key can have multiple values, so when we remove something, we need to specify both the key and the value.

The correct answer is: remove

Question **16**
Incorrect
Mark 0.00 out of 1.00
Flag question

ADT Multimap can be represented retaining unique elements and for each of them a dynamic array of associated values.

This is similar to having a simple Map, in which the values are of type dynamic array. This version is actually how you can "simulate" a MultiMap in programming languages where this container does not exist but the Map does.

For which operation would we have different types for the input parameters for these two versions?

Select one or more:

- ☐ add
- ☐ remove
- ☑ search ⊗
- ☐ size

Your answer is incorrect.

Operation add takes as parameter a key and a value for both containers. But for the MultiMap, value is actually just a value, while for Map, in this setting a value is a dynamic array, so the parameter for add would be a key and a dynamic array.

Operation remove has different parameters already on the interface of Map (just the key) and MultiMap (a key and a value).

Operation search is the same, you pass only a key as parameter.

Operation size takes no parameter, so this is the same as well.

The correct answers are:
add,
remove